AIM: Adaboost Ensemble Learning
Implement the Adaboost algorithm to create an ensemble of weak classifiers.
Train the ensemble model on a given dataset and evaluate its performance.
Compare the results with individual weak classifiers.

CODE:

```
import matplotlib.pyplot as plt
import pandas as pd
from sklearn.ensemble import AdaBoostClassifier
from sklearn.metrics import accuracy_score, classification_report
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier


df = pd.read_csv("C:/Users/hrane/Downloads/online food delivery_107.csv")
df.columns = df.columns.str.strip()
df = df.apply(lambda x: x.str.strip() if x.dtype == "object" else x)


df["Output"] = df["Output"].map({"Yes": 1, "No": 0})
X = pd.get_dummies(
    df.drop(["Output", "Feedback"], axis=1, errors="ignore"), drop_first=True
)
y = df["Output"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)


weak_clf = DecisionTreeClassifier(max_depth=1, random_state=42)
weak_clf.fit(X_train, y_train)
y_pred_weak = weak_clf.predict(X_test)
accuracy_weak = accuracy_score(y_test, y_pred_weak) * 100


ensemble_clf = AdaBoostClassifier(
    estimator=DecisionTreeClassifier(max_depth=1),
    n_estimators=50,
    random_state=42,
)
ensemble_clf.fit(X_train, y_train)
y_pred_ensemble = ensemble_clf.predict(X_test)
accuracy_ensemble = accuracy_score(y_test, y_pred_ensemble) * 100


print("=" * 50)
print(f"Individual Weak Classifier Accuracy: {accuracy_weak:.2f}%")
```


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```
print(f"AdaBoost Ensemble Classifier Accuracy: {accuracy_ensemble:.2f}%")
print("=" * 50)
print("\n--- Detailed AdaBoost Ensemble Classification Report ---")
print(classification_report(y_test, y_pred_ensemble))


plt.figure(figsize=(8, 5))

classifiers = ["Individual Weak Classifier", "AdaBoost Ensemble"]
accuracies = [accuracy_weak, accuracy_ensemble]
colors = ["#ff9999", "#66b3ff"]

bars = plt.bar(classifiers, accuracies, color=colors, width=0.5)


plt.title("Performance Comparison: Weak Learner vs. AdaBoost Ensemble")
plt.ylabel("Accuracy Percentage (%)")
plt.ylim(0, 110)
plt.grid(axis="y", linestyle="--", alpha=0.7)


for bar in bars:
    height = bar.get_height()
    plt.text(
        bar.get_x() + bar.get_width() / 2.0,
        height + 2,
        f"{height:.2f}%",
        ha="center",
        va="bottom",
        fontweight="bold",
    )


plt.tight_layout()
plt.show()
```

OUTPUT:

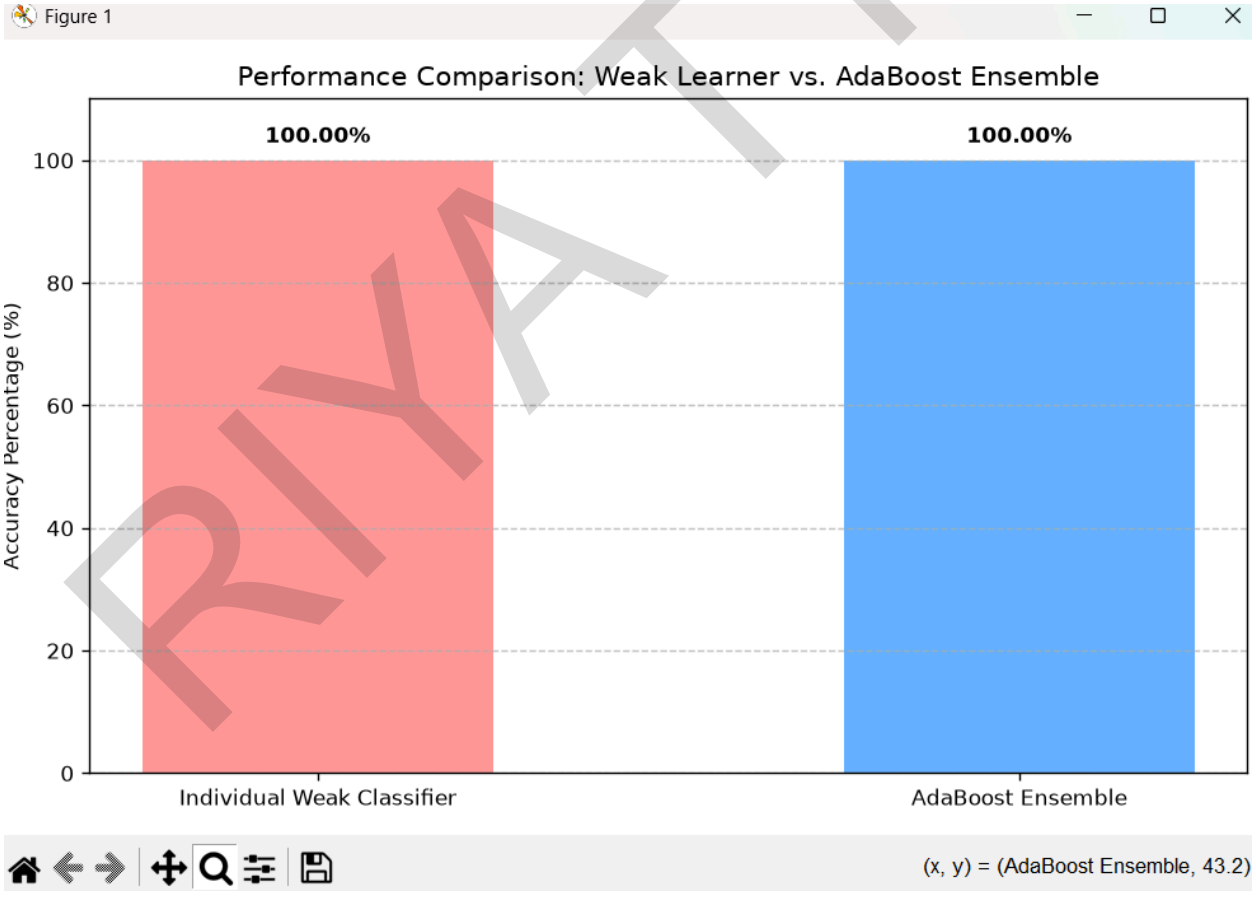
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```
------------------ RESTART: C:\Users\hrane\Documents\AI_R.py ----------------
==================================================
Individual Weak Classifier Accuracy: 100.00%
AdaBoost Ensemble Classifier Accuracy: 100.00%
==================================================

--- Detailed AdaBoost Ensemble Classification Report ---
              precision    recall  f1-score   support

           0       1.00      1.00      1.00        17
           1       1.00      1.00      1.00        61

    accuracy                           1.00        78
   macro avg       1.00      1.00      1.00        78
weighted avg       1.00      1.00      1.00        78
```



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